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ORIGINAL PAPER

Cortical Representation of Auditory Selective Attention in a Dichotic Listening Task: A Functional Near-Infrared Spectroscopy Study

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Abstract

To advance the application of functional near-infrared spectroscopy (fNIRS) in brain-computer interface (BCI) technology, we investigated cortical activation patterns associated with auditory selective attention. Using a dichotic listening paradigm, participants were presented with simultaneous music and reading sounds to the left or right ear. During fNIRS recordings, they were instructed to selectively attend to the sound attribute (music vs. reading) or the spatial location (left vs. right ear). Cortical activity differences related to attentional targets were analyzed using a two-way analysis of variance (ANOVA), with sound attribute and spatial information as factors. Our results revealed a significant main effect of the sound attribute factor across multiple measurement channels. Notably, the right parietal region exhibited consistently greater activation when attention was directed toward music compared to reading sounds. Conversely, bilateral dorsolateral prefrontal cortex (DLPFC) channels showed higher activation when participants attended to reading sounds than to music. These findings indicate that cortical activation patterns are modulated by auditory attentional states based on sound attributes. Furthermore, preliminary classification analyses achieved an accuracy of 73.7% in discriminating attentional targets (music vs. reading sounds), demonstrating the feasibility of fNIRS-based BCI applications.

Keywords BCI · FNIRS · Hemodynamics · Auditory function · Selective attention · Dichotic listening · Music · Speech

Introduction

Functional near-infrared spectroscopy (fNIRS) is a non-invasive neuroimaging technique that uses near-infrared light to measure relative changes in the concentrations of oxygenated and deoxygenated hemoglobin in the cerebral cortex (Ferrari and Quaresima 2012). The primary advantages of fNIRS over other neuroimaging modalities, such as functional magnetic resonance imaging (fMRI) and magnetoencephalography (MEG), are its lower participant burden, resistance to noise, cost-effectiveness, and portability (Koizumi et al. 2003). Due to these advantages, fNIRS has been applied in various fields, including developmental science (Homae et al. 2010; Sato et al. 2012; Taga et al. 2003) and clinical research (Monden et al. 2015; Takizawa et al. 2014; Watanabe et al. 1998). Capitalizing on these benefits, researchers have increasingly applied fNIRS in the development of brain-computer interfaces (BCIs) for bedside and daily use (Hong et al. 2018; Naseer and Hong 2015; Schudlo and Chau 2014). Furthermore, for patients with amyotrophic lateral sclerosis (ALS) or complete paralysis due to injury, who are unable to engage in voluntary physical activities (Birbaumer 2006; Kübler and Birbaumer 2008), BCIs are recognized as a crucial means for communication, offering a means to maintain and potentially enhance the quality of life (Borgheai et al. 2020). Although the present study focuses on healthy adults, this clinical context motivates our choice of task design: targeting preserved sensory and cognitive functions, such as auditory selective

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attention, is essential for developing BCIs that remain feasible even in severe paralysis.

In fNIRS-based BCI applications, selecting appropriate tasks is critical. Many BCI studies have employed motor imagery or cognitive tasks, such as mental arithmetic, which do not rely on external stimuli, allowing users to voluntarily modulate their brain activity (Hong et al. 2018; Naseer and Hong 2015). For instance, studies using motor imagery tasks have demonstrated high discrimination rates based on measurements from the motor cortex (Naseer and Hong 2013; Sitaram et al. 2007). However, for patients in the locked-in state who are unable to perform actual movements, the motor cortex may not exhibit significant activation in response to motor imagery (Conson et al. 2008; Murguialday et al. 2011). In contrast, cognitive tasks can be performed by patients even in a completely locked-in state (CLIS) (Hong et al. 2018; Kübler and Birbaumer 2008). Additionally, the prefrontal cortex (PFC) is a preferred measurement site in BCI applications not only because of its central role in attentional control and executive function, but also because hair coverage is minimal in this region, resulting in higher signal quality compared to hair-covered areas (Hong et al. 2015; Naseer and Hong 2015). Although advances in fNIRS optode technology have reduced hair-related artifacts, this anatomical advantage continues to contribute to reliable measurement. Cognitive load is better conceptualized as a continuum rather than a simple low/high dichotomy. In real-world BCI applications, tasks with lower or moderate demands are generally preferable because they minimize fatigue and maximize usability, even though highly demanding tasks may yield somewhat clearer brain signals.

Another issue is how the baseline is defined when evaluating brain signals during cognitive tasks. Because a simple resting state, during which no instructions are provided, can be uncontrolled and unstable (Schudlo and Chau 2014), this study employed an active baseline to ensure stable and reproducible brain signals during task-related analysis. This approach is consistent with prior studies that aimed to improve baseline stability by employing mild cognitive engagement instead of an uncontrolled resting condition. For instance, some studies have compared brain activity during mental arithmetic tasks using a mild cognitive task, such as counting (Naito et al. 2007) or imagining alphabet pronunciation (Shin et al. 2016) as the baseline. Ang et al. used three levels of difficulty in mental arithmetic tasks and performed a two-class classification for each condition (Ang et al. 2010). These approaches are expected to provide greater stability as they discriminate brain activity without relying on a resting state. However, the reported classification accuracy is approximately 60–80% in binary classification, which is not sufficiently high. Furthermore, higher-order cognitive tasks are considered more demanding than typical cognitive activities. While highly demanding cognitive tasks may yield clearer and more distinguishable brain signals for classification, they are not always practical for real-world BCI applications, where enabling communication requiring minimal cognitive effort is preferable.

Therefore, we are investigating a new BCI task that does not require a resting state signal or a highly demanding cognitive load to identify brain activity states. To this end, we focused on selective auditory attention for the following reasons. First, auditory sensations remain largely unaffected in ALS patients, making auditory stimuli well-suited for BCI tasks (Kübler and Birbaumer 2008; Murguialday et al. 2011). Second, selective attention—the ability to filter out irrelevant information—develops early in life (Jones et al. 2015; Ridderinkhof and Van Der Stelt 2000) and becomes habitual for many individuals. It is also frequently used in everyday life (Alain et al. 2013; Fritz et al. 2007) and may reduce individual variation in task difficulty and less cognitive burden for participants.

A representative task requiring auditory selective attention is dichotic listening, which involves presenting two distinct sounds simultaneously to the left and right ears (Westerhausen 2019). This task has been used to study bottom-up processing, such as hemispheric dominance (Della Penna et al. 2007; Hugdahl and Westerhausen 2016; Mei et al. 2020), as well as top-down control of attention (Jäncke et al. 2003; Pugh et al. 1996; Thomsen et al. 2004). Although auditory stimuli are primarily processed in the temporal regions, including the auditory cortex, the frontal region is also suggested to be involved due to the demands of selective attention. In an fMRI study, bilateral activation of the dorsolateral prefrontal cortex (DLPFC) was observed during a dichotic listening task (Pugh et al. 1996), which is attributed to the inhibition of input from one ear while focusing on the other. Other fMRI studies using dichotic listening tasks have also indicated that the activation of attentional networks within the prefrontal cortex is linked to selective attention (Jäncke et al. 2003). Additionally, an fNIRS study examined hemodynamic changes related to selective attention during a dichotic listening task (Eskicioglu et al. 2019). This study assessed brain activity under three distinct conditions: non-attentive, forced-left, and forced-right attention during a binaural dichotic listening task involving consonant-vowel (CV) syllables. Results showed significant activation in the right prefrontal cortex under forced attention conditions compared to the non-attention condition, regardless of the direction of attention (left or right). These findings suggest that activation patterns in the prefrontal cortex serve as markers of inhibition and focused attention. However, previous dichotic listening studies have primarily used simple speech or tonal stimuli. We introduced a novel

variation by pairing music with spoken text (reading) to capitalize on their distinct hemispheric processing tendencies—music being preferentially processed in right-hemisphere networks and speech in left-hemisphere networks (Limb 2006; Pastuszek-Lipińska 2025; Plakke and Romanski 2014; Perani et al. 2010; Vigneau et al. 2006). Furthermore, dichotic listening literature has reported a left-ear advantage for music and a right-ear advantage for speech (Kimura 1967; Kompus et al. 2012), reflecting both hemispheric specialization and contralateral auditory pathways. These asymmetries are not limited to the temporal cortex (Plakke and Romanski 2014); they can influence prefrontal attentional control networks, thereby providing a principled rationale for testing whether stimulus type and ear of presentation jointly modulate DLPFC activity.

Based on this framework, we hypothesized that DLPFC activation would differ depending on (i) the attended stimulus category (music vs. reading), and (ii) the ear of presentation (left vs. right), with potential interaction effects when the ear advantage aligns with the hemispheric specialization of the stimulus. This hypothesis directly links selective attention mechanisms, lateralized auditory processing, and prefrontal top-down control, thereby addressing both theoretical and practical aspects of stimulus-driven BCI design. To further evaluate the feasibility of this paradigm for BCI applications, we also performed an exploratory classification analysis to examine whether distinct patterns of DLPFC activation across attentional conditions could be used to discriminate attentional states. This analysis, conducted at the participant level using aggregated task-related features, aimed to assess the potential separability of attentional states in a practical BCI context.

Materials and Methods

Participants

Twenty-two healthy men (19–24 years old) participated in the study; all were right-handed and had no history of auditory disorders. All participants were native Japanese speakers, born and raised in Japan, and were university students or graduates from the same institution. Therefore, they shared a relatively homogeneous sociocultural and educational background, minimizing interindividual variability related to language or cultural familiarity with the auditory stimuli. Data from three participants were excluded from the analysis due to technical errors in the measurements. This study was approved by the Biotechnology Research Ethics Committee of the Shibaura Institute of Technology. Informed consent was obtained from all the participants before the experiment. The experimental protocol was performed in accordance with the "Guidelines for ethics-related problems with non-invasive research on human brain function" established by the Japan Neuroscience Society.

The target sample size for this study was determined based on practical considerations, such as participant availability, and to ensure comparability with sample sizes used in previous fNIRS studies employing similar paradigms. Although a formal a priori power analysis was not conducted, we performed post hoc power analyses using the results of the repeated-measures ANOVA (see Results) to evaluate the achieved statistical power for the observed effects.

fNIRS System

An fNIRS system (ETG-4000, Hitachi Medical Corporation, Tokyo, Japan) was used to measure brain activity signals during auditory selective attention tasks involving dichotic listening stimuli. The fNIRS system was equipped with 33 probes (17 near-infrared light sources and 16 detectors) arrayed in a 3×11 lattice pattern centered on the frontal cortex, resulting in 52 measurement points (channels). In each channel (CH) between a light source and a detector, signals for oxygenated hemoglobin (oxy-Hb) and deoxygenated hemoglobin (deoxy-Hb) were recorded at a sampling rate of 10 Hz. The source–detector distance was fixed at 30 mm, which is standard for cortical measurements.

The probe arrangement was based on the international 10–20 system, with the bottom-center probe positioned at Fpz. To estimate the measurement channel locations in Montreal Neurological Institute (MNI) space, we utilized the coordinate data published by Sato et al. (2013), which employed the same channel configuration during simultaneous fNIRS and fMRI measurements. In addition, a probabilistic registration method (Singh et al. 2005) was applied to generate 3D topographic maps, confirming adequate coverage of the DLPFC across participants. Based on this mapping, channels over the DLPFC were primarily located in the anterior–lateral areas of both hemispheres (approximately CH 2–5, 13–15, 23–26, 34–36, 45–47 in the right hemisphere; and CH 6–9, 17–19, 27–30, 38–40, 48–50 in the left hemisphere). Although the auditory cortex was not fully covered, this was not essential for the current research objectives, which focused on prefrontal activation patterns as markers of attentional control.

Dichotic Listening Task

We created dichotic listening stimuli using music and reading sounds. For the music stimuli, classical music recorded from CDs played on both traditional Japanese instruments (MUSICAL INSTRUMENT WAGAkki 01, 02,

manufactured by Nippon Television Music Corporation) and Western instruments (100 CLASSICS, manufactured by AVEX MARKETING INC.) were used. We selected standard, well-known classical pieces to minimize both linguistic interference and differences in familiarity among participants. For the reading stimuli, recordings of Japanese folk tales from CDs ("Kiku-Ehon: Mukashi-Banashi Best 100", manufactured by Pan Rolling Inc.) were used. We selected well-known Japanese folk tales that are culturally familiar and linguistically simple, so that all participants could easily understand the content without differences in comprehension or cultural background. Although no formal pre-test was conducted, informal pilot listening among laboratory members confirmed that the selected stories were easily understandable. Using Adobe Audition 2021, each sound was segmented into 20-second clips and edited to present different types of sounds to the left and right ears. There were two types of stimuli; one presented music to the left ear and reading sounds to the right ear, while the other presented reading sounds to the left ear and music to the right ear. We created 24 different stimuli for each type, resulting in a total of 48 stimuli. These stereo stimuli were delivered through earphones (SE215, Shure Inc., IL, USA). To ensure comparable perceptual intensity between the reading and music sounds, their sound pressure levels (SPLs) were adjusted based on subjective loudness matching. The output SPLs from the earphones were evaluated using a multi-environmental measuring instrument (LM-8102, Mother Tool, Nagano, Japan). Based on representative random sampling of several 20-second clips, the maximum SPL of the reading sounds within a single segment was approximately 67 dB, and that of the music sounds was approximately 66 dB. The minimum SPL values were not reported because short pauses in the reading stimuli produced transient near-silent periods, which do not reflect the perceived loudness of the auditory stimuli. Therefore, these measurements are presented as representative estimates of the typical maximum levels across all stimuli.

For the dichotic listening task, there were two types of sessions with different instructions. In the "attribute instruction" session, participants were instructed to pay attention to either the "music" or the "reading" sounds. In the "spatial instruction" session, they were instructed to focus on the "sound presented to the left ear" or the "sound presented to the right ear." These instructions were displayed on a screen in front of the participants for 5 s before the task began. Notably, participants always attended to the sound attribute, as their performance was evaluated based on responses to a simple two-choice question related to the sound attribute in both sessions. The questions were "What is the title of this story?" when they focused on the reading sound and "What is the style of this music?" when they focused on the music sound. Thus, we defined four conditions: music presented to the left ear (Music-Left condition), reading sounds presented to the right ear (Reading-Right condition), reading sounds presented to the left ear (Reading-Left condition), and music presented to the right ear (Music-Right condition), based on the sound attribute of the attended target, regardless of the instruction type (Fig. 1(a)).

In each session, three trials for each condition were presented in random order, resulting in a total of 12 task trials per session. Each participant completed four sessions: two attribute instruction sessions and two spatial instruction sessions, yielding a total of 48 task trials. The experimental sessions were spread across two days to minimize the influence of experimental fatigue and to enhance result reproducibility.

Experimental Procedure

Participants attended two experimental days, separated by one week. On the first experimental day, they were informed about the study and asked to complete a consent form and an information sheet to record details such as age, gender, and handedness. Four practice trials were conducted before the actual measurements began.

On each experimental day, participants sat in a chair in a sound-attenuated chamber while wearing the fNIRS probe cap. They then completed an attribute instruction session and a spatial instruction session in a counterbalanced order. As mentioned above, each session consisted of 12 task trials (Fig. 1(b)). The duration of the rest period between trials was set to 20 s to ensure sufficient recovery of the hemodynamic response before the onset of the next task block. In previous studies using basic sensorimotor paradigms, most participants' hemodynamic waveforms returned to baseline within 10–15 s after stimulus offset, and these responses were shown to be highly reproducible (Sato et al. 2006). Furthermore, the best-practice recommendations for fNIRS experimental design (Yücel et al. 2021) illustrate a 15-second inter-block interval as a representative example. Therefore, the present study adopted a 20-second rest period, which falls within the standard range commonly used in fNIRS block designs and provides a conservative buffer to ensure full baseline recovery. The time sequence of each task trial was as follows: 20 s of rest, 5 s of attention instruction, 20 s of task performance, and up to 5 s of response to a question followed by an additional 20 s of rest. Each session lasted approximately 11 min, and two sessions were conducted per day over two experimental days (approximately one week apart). Thus, the total active task time per participant was within 30 min per day. This design minimized participant fatigue and ensured that the total measurement duration remained within the range typically

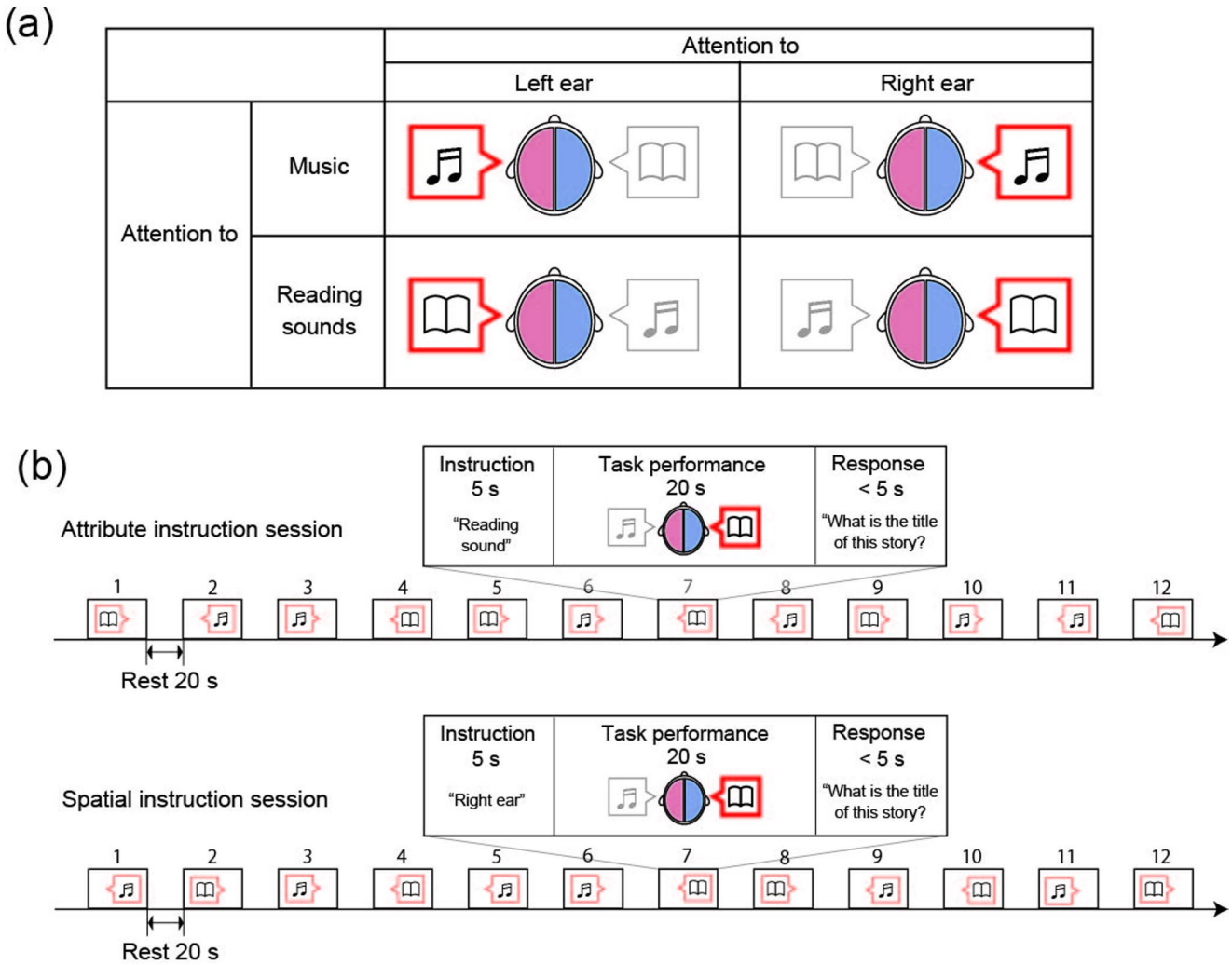

Fig. 1 Experimental design and attentional conditions. (**a**) Two types of auditory stimulus configurations and the resulting four attentional conditions. The "suitable" configuration consisted of music presented to the left ear and reading sounds to the right ear, whereas the "unsuitable" configuration presented reading sounds to the left ear and music to the right ear. Four distinct attentional states were measured by directing participants' focus toward one of the two sounds, yielding the following conditions: music in the left ear, music in the right ear, reading in the left ear, and reading in the right ear. In the schematic, the ♬ and book symbols represent the positions of the music and reading sounds, respectively. Letters within the figure indicate the sound attribute (music or reading) and spatial information (left or right ear) of the attended target. (**b**) Schematic illustrating the time sequence of the two session types—attribute instruction and spatial instruction—and the detailed structure of a single task trial. Each session consisted of 12 task trials, each comprising an instruction phase (5 s), a task performance phase during which participants attended to the designated target sound, and a response phase (within 5 s) in which they answered a two-choice question regarding the attended sound attribute. The two session types were conducted in a counterbalanced order within a single recording day. Each participant completed four sessions in total across two recording days

used in fNIRS block-design studies. During the rest periods, participants were instructed to relax, clear their minds, and fixate on a cross displayed at the center of the screen while avoiding head movement. In the attention instruction phase, the screen indicated where participants should direct their selective attention—toward the music, reading sounds, left ear, or right ear. During task performance, participants focused their attention on the instructed target. In the response phase, participants answered a two-alternative question related to the sound attribute by pressing a key. Questions included prompts such as "What is the title of this story?" or "What is the style of this music?"

Behavioral analysis focused on participants' responses to these simple questions to confirm that their attention was correctly aligned with the intended stimulus. Response accuracy was computed for each participant and a repeated-measures analysis of variance (ANOVA) was performed, factoring in both the sound attributes (music or reading sounds) and the spatial information (left or right ear) of the attended stimulus. Only responses within the 5-second

window, the maximum duration allocated for the question, were accepted.

Preprocessing of fNIRS Signals

The fNIRS signals were preprocessed using the Platform for Optical Topography Analysis Tools (POTATo) (Sutoko et al. 2016) in MATLAB (MathWorks). First, the light attenuation data, represented by the voltage signal, were converted into oxy- and deoxy-Hb signals based on the Beer–Lambert law (Maki et al. 1995). Second, a band-pass filter (0.01–0.8 Hz) was applied to both Hb signals to eliminate low-frequency drift/oscillation and high-frequency noise components (Kamran et al. 2016; Naseer and Hong 2015).

Subsequently, the signals were divided into functional and systemic components using the hemodynamic modality separation method (Yamada et al. 2012), which separates neural-related hemodynamic responses from global physiological oscillations. Only the functional signals were retained for further analysis to minimize physiological noise contamination. Furthermore, since the paradigm involved multiple repetitions of each attentional condition, the subsequent trial-averaging process effectively reduced uncorrelated physiological noise and transient artifacts, enhancing the signal-to-noise ratio of task-related responses. Their method separates the signals based on their physiological origins, assuming that the functional components of oxy- and deoxy-Hb signals show an inverse correlation. This procedure also inherently reduces the influence of outliers and physiological noise unrelated to task-induced activation. Compared with commonly used filtering and preprocessing techniques summarized by Herold et al. (2018), this modality separation approach offers a practical advantage because it does not require additional hardware such as short-separation channels while effectively isolating functional components from systemic interference. After applying this method, the oxy- and deoxy-Hb functional signals were statistically identical, indicating that they contained equivalent functional information. Both parameters were examined, but as they produced statistically identical results, only the oxy-Hb signal is reported here to avoid redundancy. The continuous-time data of the oxy-Hb signals recorded during each session were segmented into 50-second task blocks to calculate the block average. Each block consisted of a 6.0-second pre-task period (1 s of rest and 5 s of attention instruction), a 20-second task period, and a 24-second post-task period (5 s of response and 19 s of rest). Baseline correction was applied using the 1-second rest signal immediately preceding the attention instruction for each block. This duration was selected to (1) minimize fluctuations carried over from the preceding task and response phases, and (2) maximize the post-task resting period available for subsequent analyses. Although a 1-second baseline may appear short compared with more conventional 5-second periods, our previous studies have shown that it yields stable activity waveforms in similar experimental paradigms (Sato et al. 2012; Aoki et al. 2011).

Statistical Analysis

The brain activity values used for the analysis were computed as follows. To explore potential temporal changes in task-related activation, two-time windows were defined as analysis periods to explore the common activation associated with the dichotic listening task across all attentional conditions: the first half (0.1–10.0 s) and the second half (10.1–20.0 s) of the stimulus presentation period. The first half was expected to capture the initial phase of attention engagement, whereas the second half might involve sustained attention or possible habituation effects. Initially, the mean oxy-Hb signal for each attentional condition in both analysis periods was calculated. Using these mean values from individual trials, z-values were computed as the common brain activity value by dividing the mean values across trials by the corresponding standard deviation (Eqs. 1–3) (Sato et al. 2011).

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

$$SD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2} \tag{2}$$

$$z = \frac{\bar{x}}{SD} \tag{3}$$

Using the common brain activity values, t-tests were conducted to determine which channels were significantly activated by the dichotic listening tasks in both the first and the second halves of the stimulus presentation period. All 52 channels covering the prefrontal region were included in this exploratory analysis to examine the overall spatial distribution of activation patterns. The threshold for p-values was corrected for multiple testing using a permutation test (Nichols and Holmes 2002). Specifically, forty-eight 26-second windows were randomly sampled from the oxy-Hb signals across four sets of 12-trial measurements, where band-pass filtering and hemodynamic modality separation methods had been applied. The 26-second window corresponds to one task period, including 1 s of pre-task and 20 s of post-task period, with no overlap between windows. Subsequently, the z-values for each CH were obtained in the same manner as in the real analysis described earlier. A t-test was then conducted on these z-values across participants for

each CH, and the maximum t-value among the 52 CHs was recorded for each iteration. This process was repeated 1000 times, and the t-value corresponding to the upper 5% of the maximum t-values across all simulations was used as the threshold value, corrected for multiple comparisons. Channels with t-values higher than the threshold obtained from the permutation test were defined as activated channels (ACT-CHs) in both the first and second halves of the task. To maximize sensitivity while maintaining reliability, we selected the analysis period that best captured robust task-related activation for subsequent analyses. The rationale and specific results supporting the choice of the first-half window are presented in the Results section. Both analysis windows were examined in an exploratory manner, and the activation patterns for each window are reported in the Results section to empirically justify the selection of the first-half window. The choice of a 10-second division is also consistent with previous neuroimaging studies, as both BOLD fMRI and fNIRS share the temporal characteristics of the neurovascular hemodynamic response, and several fMRI studies have employed analysis windows of approximately 10 s to characterize activation patterns (e.g., Fox et al. 2005; Wen et al. 2019). This segmentation thus allowed us to capture both the initial rising phase and the later, potentially habituated phase of the task-evoked hemodynamic response, providing a balanced temporal resolution for detecting dynamic changes in activation.

Next, z-values were calculated for each attentional condition—Music-Left, Reading-Left, Music-Right, and Reading-Right—in the ACT-CHs for each participant to perform a repeated-measures ANOVA. This ANOVA was based on a 2 × 2 repeated-measures factorial design, with the primary factors being sound attributes (music versus reading sounds) and spatial information (left versus right ear) corresponding to the stimuli that directed attention. In this test, the partial eta squared value was calculated as the effect size.

All trials, including those in which participants either failed to respond within the 5-second window or responded incorrectly (19 out of 1,056 trials; 1.8%), were included in the statistical analyses. These cases were regarded as unavoidable noise inherent to human behavior, occurring even when participants attended to the target stimulus.

Table 1 EBehavioral results showing average accuracy and standard error for simple questions on stimuli to which selective attention was directed, across four attention conditions

Accuracy [%]	Left	Right
Music	96.9 ± 1.55	97.6 ± 1.89
Reading sounds	98.7 ± 0.697	97.8 ± 1.05

Results

Behavioral Analysis

Basic behavioral data are shown in Table 1, indicating that participants responded to the simple question after the task with an average accuracy of more than 96% across all conditions. Repeated-measures ANOVA did not reveal any significant main effects or interactions between sound attributes and spatial information (main effect of sound attributes: $F(1,18) = 0.443$, $p = 0.514$, $\hat{\eta}_P^2 = 0.0150$; main effect of spatial information: $F(1,18) = 0.00617$, $p = 0.938$,, $\hat{\eta}_P^2 = 0.0000106$; interaction: $F(1,18) = 2.660$, $p = 0.120$,, $\hat{\eta}_P^2 = 0.0105$).

General Activation Channels during the Dichotic Listening Task

In the permutation test, the results of 1000 simulations for each analysis period yielded t-value thresholds of 3.23 for the first half and 3.26 for the second half. Using these thresholds, ACT-CHs were determined for both analysis periods, revealing eighteen ACT-CHs in the first half (Fig. 2, left) and nine ACT-CHs in the second half (Fig. 2, right) around the DLPFC. All ACT-CHs identified in the second half overlapped with those in the first half, indicating that the first-half window captured the most robust task-related activation. Consistent with the rationale described in the Methods section, only the ACT-CHs from the first half were therefore used for subsequent statistical analyses.

Cortical activation pattern depending on selective attention.

We conducted a repeated-measures ANOVA to examine the cortical activation patterns depending on auditory selective attention to sound attributes (music vs. reading sounds) or spatial information (left vs. right ear) in the ACT-CH (Fig. 3). Significant main effects of sound attributes were detected in three channels (CH 2, CH 25, and CH 28) (Fig. 3(a)), and significant interaction effects were found in CH 13 (Fig. 3(c)). In addition, no significant effects of spatial information were observed.

The significant channels that showed a main effect in the ANOVA were examined in detail (Fig. 4, red squares). In CH 2 (BA6), which showed the main effect of sound attributes, higher activity was observed when listening to music compared to reading sounds (first half: $F(1,18) = 6.07$, $p = 0.0240$, $\hat{\eta}_P^2 = 0.123$). Conversely, we found the opposite results in CH 25 and CH 28 (BA46), which also showed the main effect of sound attributes: attention to reading sounds induced higher activation than attention to music (CH 25:

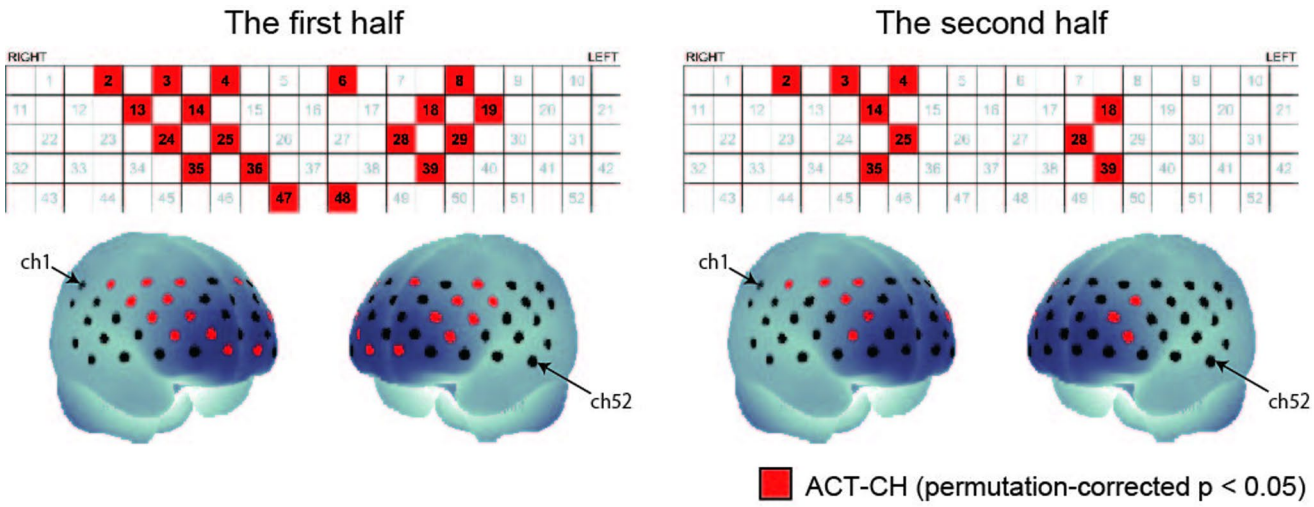

Fig. 2 Distribution of activated channels (ACT-CHs) during the selective auditory attention task involving dichotic listening stimuli. The left figure presents the results for the first half, and the right figure presents the results for the second half of the analysis period. In these figures, CHs whose t-values surpassed the threshold determined through simulation are denoted by red markers

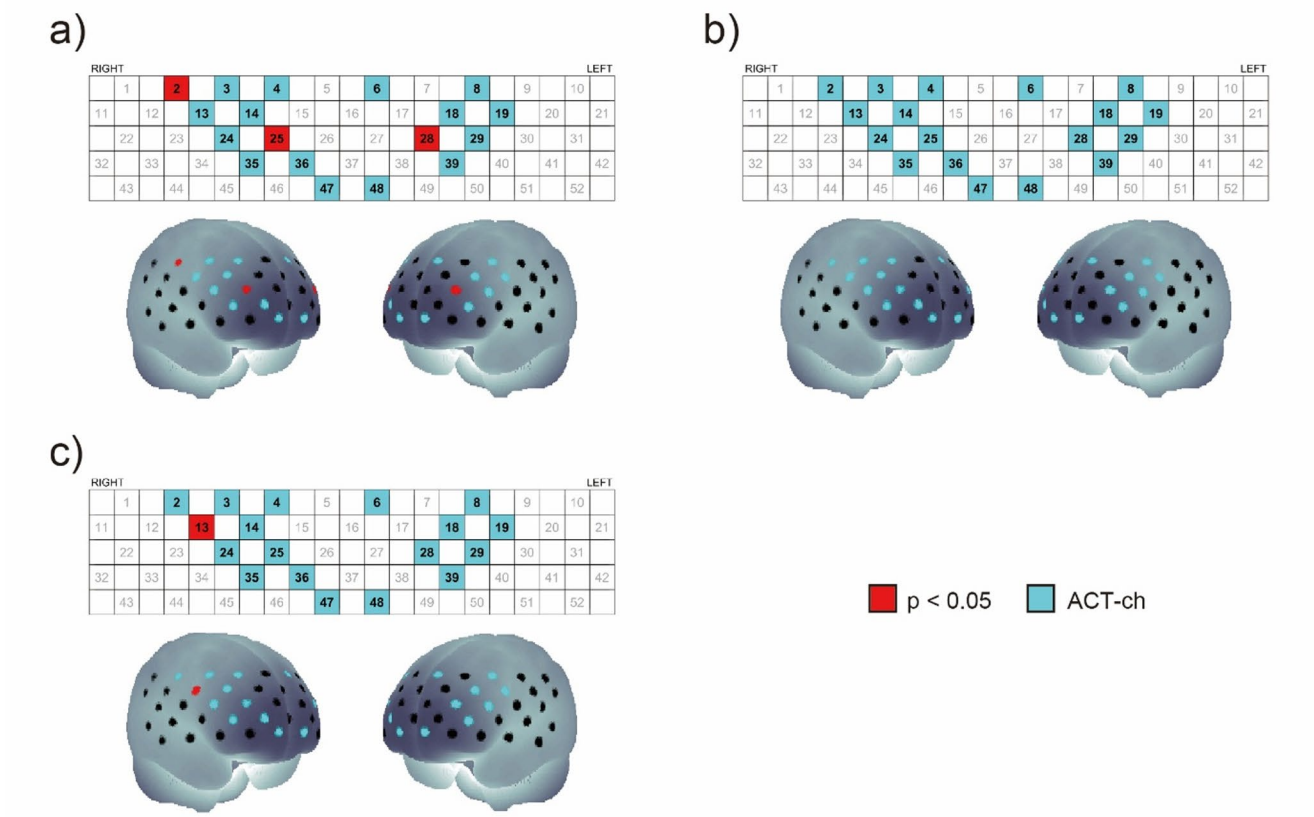

Fig. 3 Results of ANOVA (sound attribute × spatial information). The analysis was carried out for each ACT-CH. (**a**) Results in the main effects of sound attributes; (**b**) results in main effects of spatial information; and (**c**) results in the interaction. Positions shown in light blue represent ACT-CH, while positions shown in red signify channels representing statistical significance in either main effects or interactions

F(1,18) = 6.90, p = 0.0171, $\hat{\eta}_P^2$ = 0.104; CH 28: F(1,18) = 6.60, p = 0.0193, $\hat{\eta}_P^2$ = 0.098).

A few channels showed a significant interaction between sound attributes and spatial information in the ANOVA (Fig. 4, yellow squares). CH 13 exhibited a significant interaction (F(1,18) = 5.11, p = 0.0364, $\hat{\eta}_P^2$ = 0.068). Subsequent post hoc tests (corrected using Tukey's honest significant difference (HSD) method) revealed higher activation in the

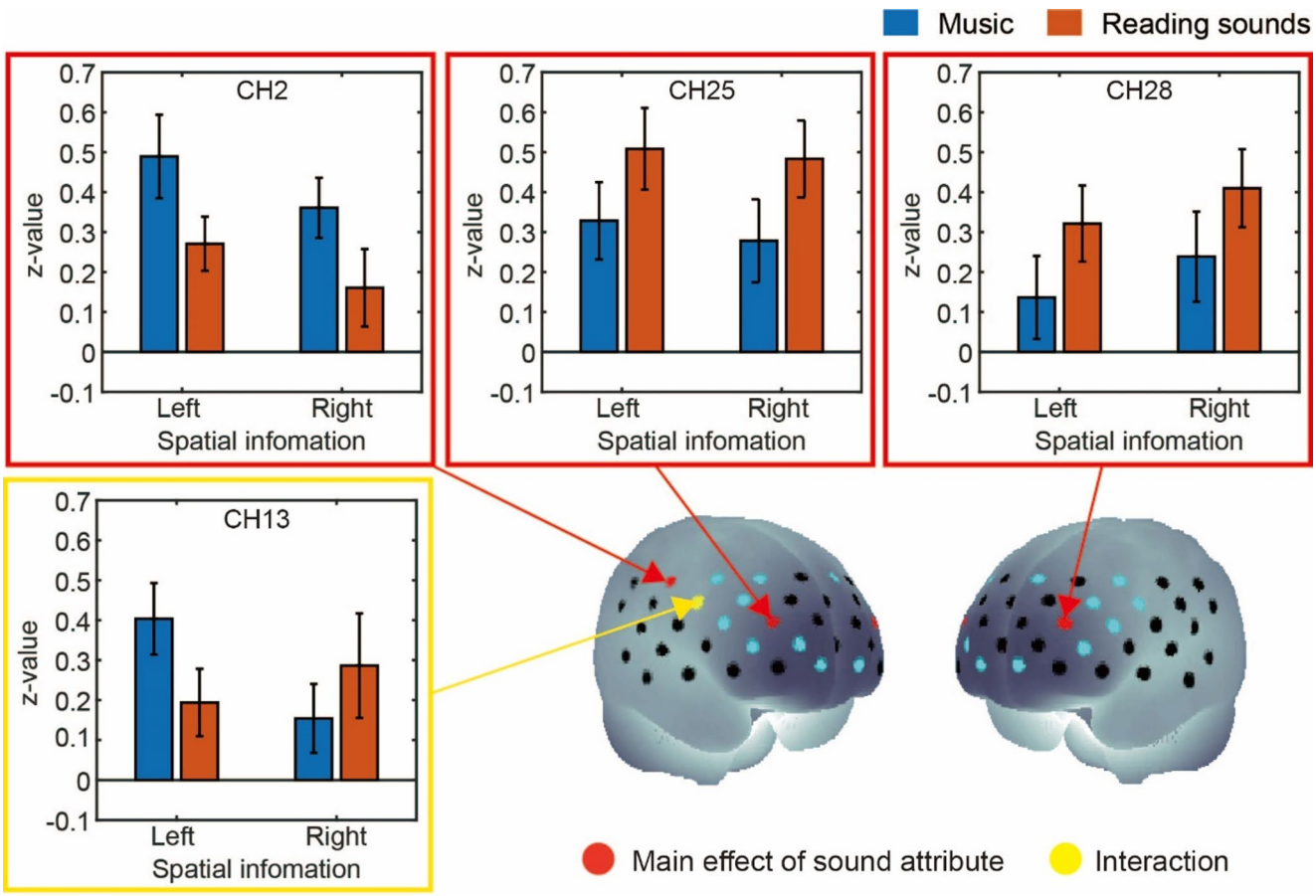

Fig. 4 Detailed results of the ANOVA conducted for the first half of the analysis period. In the brain map, light blue represents ACT-CHs, red designates channels with significant main effects related to sound attributes, and yellow signifies channels with significant interaction effects. In the bar graphs, blue and orange bars show the attentional state of focusing on music and reading sounds, respectively. Spatial information is shown on the horizontal axis of the graph. The activation level (mean z-value) for each attentional condition is presented in the bar graphs, with error bars indicating the standard error

Music-Left condition than in the Music-Right condition ($p = 0.0185$). The graph for CH 13 also shows distinct high and low activity conditions. The high activity conditions were Music-Left and Reading-Right, while the low activity conditions were Music-Right and Reading-Left, suggesting that Music-Left (Reading-Right) stimuli elicited higher activation than Music-Right (Reading-Left) stimuli.

We performed *a* posteriori power analyses using the results of the repeated-measures ANOVA and the software G*Power 3.1.9.7. First, the partial eta squared (η^2) was calculated for each channel showing a main effect to determine the effect size. Power calculations with alpha set to 0.05 and sample size set to 19 yielded power values of 0.87, 0.80, and 0.77 for CH 2, CH 25, and CH 28, respectively.

Exploratory Classification Analysis

In addition to the pre-planned analyses, we performed an exploratory classification analysis to assess whether brain activity states could be identified from fNIRS signals using labels indicating music versus reading sounds. Channels showing a significant main effect of sound attributes in the ANOVA were used as features, with the aim of testing whether accurate classification could be achieved using a compact, cost-effective set of channels suitable for practical BCI implementation.

We acknowledge that using the same dataset for both feature selection and classification can raise concerns about circular analysis ("double dipping"), which can lead to optimistically biased accuracy estimates. While group-level statistical significance reflects consistent differences across participants, it does not necessarily imply high classification performance at the individual level (Hebart and Baker 2018; Haufe et al. 2014). Therefore, the present classification should be considered as complementary evidence to the group-level results rather than as definitive proof of discriminability. This analysis was intended purely as a preliminary step to explore feasibility, and future work should employ independent feature selection procedures or external validation datasets for more rigorous evaluation.

Table 2 Classification accuracy of three models utilizing Support Vector Machine (SVM) for binary classification of attention to music and reading sounds

Model	Number of channels	Accuracy [%]
All-CH	52	50.0
ACT-CH	18	65.8
Effective-CH	3	73.7

Machine learning was performed using the Classification Learner application in MATLAB. The "Optimized Support Vector Machine (SVM)" was selected as the classification algorithm, with hyperparameters automatically tuned via Bayesian optimization over 100 iterations and evaluated using fivefold cross-validation. Four hyperparameters were optimized: kernel function, box constraint level, kernel scale, and data standardization.

Classification performance was assessed across participants, where the dataset consisted of brain activity values for all four attentional conditions of each participant. These values were labeled according to their corresponding attentional states for sound attributes, either as music (Music-Left, Music-Right conditions) or reading sounds (Reading-Left, Reading-Right conditions). Three distinct classification models were evaluated: (1) the all-CH model, using all channels; (2) the ACT-CH model, using the 18 channels (ACT-CHs) showing significant activation in the main task analysis; and (3) the effective-CH model, using the three channels with significant main effects of sound attributes. The classification results are presented in Table 2, with accuracies of 50.0%, 65.8%, and 73.7% for the all-CH, ACT-CH, and effective-CH models, respectively. In both the ACT-CH and effective-CH models, the results significantly surpassed the chance level of 50%.

Discussion

We examined cortical activation patterns depending on auditory selective attention in dichotic listening stimuli consisting of music and reading sounds. We found distinct activation patterns related to the attended sound attributes in cortical regions, including the DLPFC.

In the dichotic listening task, a simple question about the auditory stimuli was asked to confirm participants' attention to the targeted sounds. The behavioral data, represented by the accuracy of the responses, was greater than 96%, on average across all conditions. This confirms that, across all conditions, participants' attention was reliably directed toward the appropriate stimuli with high accuracy. Furthermore, the ANOVA indicated no significant main effects or interactions involving attentional conditions, confirming that participants were able to correctly focus on the target stimulus regardless of differences in sound attributes or spatial information.

Significant activation (ACT-CH) was observed in regions centered around the DLPFC during both analysis periods of the dichotic listening task. The wider spatial extent of activation in the first half compared with the second half may reflect the temporal profile of the hemodynamic response, such as earlier peak latency, and may also be related to the initial cognitive demands of the task and attention setting. The DLPFC, in conjunction with the parietal cortex, forms a network responsible for the top-down coordination of sensory processing (Gazzaley and Nobre 2012). This region has been identified as playing a critical role in detecting and maintaining attention, as well as in working memory (WM) related to auditory stimuli (Kane and Engle 2002; Okada et al. 2010; Plakke and Romanski 2014). Indeed, research focusing on dichotic listening stimuli and selective attention has consistently observed activation in frontal regions, especially in the DLPFC (Jäncke et al. 2003; Pugh et al. 1996; Thomsen et al. 2004). On the other hand, we did not observe any significant activation in the temporal cortex, whereas previous studies have reported activation in this region during dichotic listening tasks (Jäncke et al. 2003; Sato et al. 1999). Regarding this inconsistency, it is possible that the present task did not require as much attention as in previous studies. In the earlier fNIRS study, participants were instructed to continuously respond to whether a tone presented every 2 s was delivered to the right or left ear, requiring sustained attention (Sato et al. 1999). Similarly, in the fMRI experiment, participants were asked to judge the pitch of a pure tone presented every second (Jäncke et al. 2003). In contrast, the task in the present study did not require as high a level of continuous attention, as participants were instructed to focus on the stimulus but were only required to respond to a simple task after a 20-second interval. This finding suggests that a high level of sustained attention is necessary to maintain significant activation in the auditory cortex.

ACT-CH exhibited more consistent activation in the right than in the left hemisphere. This pattern possibly reflects selective attention-driven activation of the right frontoparietal network, which is activated in association with auditory spatial and selective attention (Kong et al. 2014; Lee et al. 2014; Li et al. 2012), as verified during attentional tasks involving dichotic listening stimuli (Jäncke et al. 2003; Westerhausen et al. 2010). Beyond simple attentional processes, this network is also implicated in a range of functions, including motor planning, mental imagery, mental rotation, and working memory (Ptak et al. 2017). As our task required both selective attention to dichotic listening stimuli and retention of stimulus information, it is plausible

that the right frontoparietal network was predominantly activated.

The main effect of the sound attribute was found in CH 2 (BA6). This region displayed heightened activation during attention to music compared to reading sound attention. BA6 corresponds to the premotor and supplementary motor areas. Previous studies on auditory perception and imagery have identified activation in this region across various sounds, nonverbal utterances, and music (Bengtsson et al. 2009; Gordon et al. 2018). This activation persisted even in tasks devoid of a motor component (Lima et al. 2016). Furthermore, the supplementary motor area (SMA) and pre-SMA are hypothesized to be activated by the structural elements of music, such as rhythms and beats (Gordon et al. 2018). Compared to fMRI, which has been used in many previous studies, fNIRS can be conducted in a quiet environment, allowing participants to hear the structural features of music, which may result in BA6 activation.

In contrast, higher activation of attention to reading sounds than to music was found in CH 25 and CH 28, which showed significant main effects of attribute information. The DLPFC, which corresponds to these channels, is integral to the WM and plays a pivotal role in information maintenance. In this study, participants were required to retain information about the attentional target for subsequent simple questions. Specifically, when the target was music, the participants answered questions identifying the type of instrument being played. In contrast, participants were queried about the title when the attentional target was a narrated story. It is worth noting that, whereas questions about music sought to distinguish between conventional and Japanese instruments, those pertaining to reading required participants to recall distinct titles for each story. This suggests a potentially greater WM load for attention to reading sounds than to music. Consistent with this finding, previous studies have indicated that the DLPFC is activated under a heightened WM load (Ayaz et al. 2012; Huang et al. 2013; Power et al. 2012). Furthermore, given that the DLPFC is also implicated in verbal WM tasks (Crottaz-Herbette et al. 2004; Wei et al. 2004), it is plausible that this, combined with inherent task demands, resulted in the observed higher attentional activity to reading sounds.

We found significant interactions in CH 13, where the Music-Left and Reading-Right conditions generally exhibited higher activation, while the Music-Right and Reading-Left conditions showed lower activation. The Music-Left and Reading-Right conditions correspond to an identical stimulus, where music is directed to the left ear and readings to the right, whereas the Music-Right and Reading-Left conditions correspond to another stimulus, in which the auditory configuration is reversed. Thus, it is possible that the activity was not generated by attention to a target sound but was dependent on the type of dichotic listening stimulus presented. Previous studies have indicated hemispheric dominance for musical and linguistic stimuli, with the left ear advantage (LEA) associated with musical stimuli and the right ear advantage (REA) associated with linguistic stimuli (Kimura 1967; Kompus et al. 2012; Majidpour et al. 2022). When such dominance is manifested, detection becomes more rapid and precise. These ear advantages can be attributed to two factors: hemispheric dominance and the architecture of the auditory processing pathways. Music is predominantly processed in the right hemisphere (Limb 2006; Perani et al. 2010), whereas linguistic information from reading is processed primarily in the left hemisphere (Dehaene-Lambertz et al. 2002; Vigneau et al. 2006). For dichotic listening stimuli, input from the contralateral ear inhibits the ipsilateral pathway, making the contralateral auditory pathway more dominant, which transmits information more rapidly and abundantly than the ipsilateral pathway (Kimura 1967). Within the context of this study, the Music-Left and Reading-Right conditions were aligned with LEA and REA, respectively, whereas the Music-Right and Reading-Left conditions did not. These insights suggest the existence of regions that, apart from those activated based on the attention target's information, might also be activated according to the arrangement of dichotic listening stimuli. Nevertheless, our findings do not entirely conform to these brain regions, highlighting the need for further investigation. Significant distinctions between music and language typically emerge in temporal regions, such as the auditory cortex, rather than in frontal regions, necessitating further validation in this domain. One possible reason for the absence of significant effects in the temporal channels is that the optode arrangement in this study covered only part of the auditory cortex, with limited coverage of higher-order auditory areas, because it was optimized to target the dorsolateral prefrontal cortex (DLPFC) in line with our primary aim of examining frontal activation related to selective attention. In addition, the DLPFC may exhibit robust modulation in this task because it is critically involved in top-down attentional control, integrating auditory input with task goals and suppressing irrelevant streams, regardless of whether the primary sensory response in the auditory cortex reaches statistical significance. This is consistent with prior neuroimaging findings showing that prefrontal activation can occur even when sensory cortical responses are weak or undetectable, provided that the task engages executive control processes. Future studies incorporating probe arrangements that cover both frontal and temporal regions would help clarify the relationship between attentional control networks and auditory sensory processing.

The exploratory cross-participant classification analysis indicated that attentional states for different sound attributes

could be discriminated from fNIRS signals with accuracies above the chance level when using a subset of channels showing robust group-level effects. In particular, the effective-CH model, consisting of only three channels, achieved the highest accuracy (73.7%), suggesting the potential feasibility of developing compact and cost-effective fNIRS-based BCI systems for auditory selective attention tasks. For BCIs, especially those based on EEG or fNIRS, classification accuracies in the range of 0.70–0.90 are not uncommon (Huggins et al. 2011; Perelmouter and Birbaumer 2000), indicating that our result lies within the functional range for practical use, particularly in binary-choice communication systems where some degree of error can be tolerated. Nevertheless, surveys of potential end-users, such as individuals with amyotrophic lateral sclerosis (ALS), show that many prioritize accuracies of at least 90% as a key requirement (Huggins et al. 2011). Thus, while our current accuracy is encouraging, further improvements toward this target will be necessary for widespread adoption.

One important limitation is the potential risk of circular analysis ("double dipping"), as the same dataset was used for both feature selection and classification. This approach can lead to optimistically biased accuracy estimates. It is also important to note that statistical significance at the group level does not necessarily imply high classification performance at the individual level (Hebart and Baker 2018; Haufe et al. 2014). Indeed, Haufe et al. (2014) demonstrated that features showing high statistical significance are not guaranteed to improve classification accuracy, especially in high-dimensional neuroimaging data. In addition, the relatively small sample size ($n = 19$) limits the generalizability of the classification results. Therefore, the classification analysis in this study should be regarded as exploratory in nature, intended to evaluate the feasibility of distinguishing attentional states from DLPFC activation patterns rather than to establish a generalizable predictive model. Therefore, we consider our classification analysis as providing complementary evidence to the statistical results by directly testing whether the observed effects can be used for prediction. Future studies should validate the approach using independent feature selection, larger and more diverse participant samples, and external datasets to rigorously assess generalizability. For this purpose, it will be important to identify features that can reduce or tolerate trial-to-trial variability among participants and investigate more optimal learning methods (Aydin 2020; Hong et al. 2018; Kwon and Im 2021; Petrantonakis and Kompatsiaris 2018).

Beyond the considerations specific to the classification analysis, this study had some additional limitations. First, the effect of gender differences should be considered, although we restricted the gender of the participants to males, primarily because of potential gender differences in selective attention and auditory processing. For instance, selective attention and functional cerebral networks in women are influenced by menstrual cycles (Thimm et al. 2014). Furthermore, gender differences have been observed in selective attention responses to dichotic listening stimuli (Voyer 2011). This factor warrants careful consideration when evaluating the potential applicability of our findings to female participants. Second, we discuss the application with actual users and patients with CLIS. In this study, experiments were conducted on adult male subjects without CLIS, and our results suggest the possibility of using auditory selective attention to classify brain activity. However, it is unclear whether the same results can be obtained when applied to patients with CLIS. Several studies have shown that auditory and cognitive functions are maintained in patients (Kübler and Birbaumer 2008; Murguialday et al. 2011) with CLIS and that fNIRS-based BCI with mental tasks can be applied to patients with CLIS (Borgheai et al. 2020; Naito et al. 2007). However, CLIS patients may show different brain activity trends than healthy subjects (Chenji et al. 2016; Khalili-Ardali et al. 2021; Naito et al. 2007). Therefore, it is necessary to examine whether the proposed BCI task will be useful for patients with CLIS. Third, we did not apply explicit rejection criteria for individual noisy channels, as our trial-averaging approach was expected to substantially reduce random noise in the present paradigm. However, this strategy may not fully address all sources of poor channel quality, and future work should incorporate systematic channel quality assessment and rejection procedures to align with current best practices. Finally, the effects of "habituation" to the stimuli and task procedures should be considered. In this study, 12 consecutive trials were conducted. Because all dichotic listening stimuli were different, it is unlikely that there was complete habituation to the auditory content. However, the use of the same type of question ("What is the title of this story?" or "What is the style of this music?") across trials may have contributed to the ceiling behavioral accuracy, raising the possibility that participants primarily attended to the stimuli in the initial trials to obtain the necessary information. Such a strategy could potentially lead to greater fNIRS responses in the first half compared to the second half. In addition to stimulus-related habituation, procedural habituation associated with the temporal structure of the task should also be considered. Although the overall sequence of each trial was consistent, the inter-stimulus interval was not strictly fixed because the response period allowed up to 5 s, resulting in small variations in the onset-to-onset timing across trials. This variability may have reduced complete temporal predictability; however, the general structure of the task remained highly regular, and participants may still have formed expectations about the trial flow. Such procedural predictability

could contribute to reduced attentional engagement in later portions of the task and may partially explain the weaker hemodynamic responses observed in the second half of the task window. Nevertheless, as each trial presented different auditory content and the questions were only revealed after stimulus offset, participants were still required to attend to the entire stimulus to answer correctly. In actual BCI use, habituation to stimuli and tasks may occur over extended sessions, and continuous task practice is known to decrease the range and intensity of activation in attention and control areas, including the prefrontal cortex, anterior cingulate cortex, and posterior parietal cortex (Kelly and Garavan 2005). Further studies are required to investigate these issues and to develop strategies to mitigate potential habituation or early-trial attention biases.

In conclusion, we investigated the relationship between the control of auditory selective attention and brain activity and attempted to classify brain activity using common trends among participants. In investigating brain activity for auditory selective attention, we used dichotic listening stimuli comprising both music and reading sounds and found significant activation, mainly in the DLPFC. This activation was particularly pronounced in the right hemisphere, which may correspond to the right frontoparietal network associated with attentional control. BA6 exhibited increased activity during attention to music, whereas BA46 was more active during attention to reading sound. Focusing on channels that exhibited activation dependent on the attentional target, we attempted an exploratory cross-participant classification, which achieved a promising classification accuracy of 73.7%. This result suggests the potential applicability of fNIRS-based BCIs that leverage auditory selective attention control. In future studies, we aim to adapt these findings to individual properties to further enhance accuracy.

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Data Availability Raw data in support of the conclusions of this article will be available from the authors upon request.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval The Biotechnology Research Ethics Committee of the Shibaura Institute of Technology approved this study (Receipt No. 21−012). Informed consent was obtained from all the participants before the experiment. The experimental protocol was performed in accordance with the "Guidelines for ethics-related problems with non-invasive research on human brain function" established by the Japan Neuroscience Society.

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